

Roll No.....

SET-I

Mid Semester Examination, March 2025

Name of the Program: B.Tech. Biotechnology

Semester: VI

Name of the Course: Biosensor and Bioelectronics

Course Code: TBT604b

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing *any one of the sub questions*.
(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Define a biosensor and explain its key components. How do biomolecular ingredients contribute to biosensor functionality? Also, describe the evolution of biosensors, highlighting key milestones.	
OR		
(b)	Define an electronic tongue and explain its working mechanism. Discuss its applications in the food and beverage industry.	
Q2	(10 marks)	CO2
(a)	Explain the working principles and applications of wearable biosensors. How do they contribute to real-time health monitoring? Additionally, describe the mechanisms of the electronic nose in detail.	
OR		
(b)	Discuss the major challenges in biosensor development and how advancements in biotechnology are addressing them. Compare point-of-care biosensors with traditional laboratory-based biosensors.	
Q3	(10 marks)	CO3
(a)	Explain the principles and applications of optical biosensors. How do they differ from electrochemical biosensors? Compare label-free and label-based biosensors, discussing their advantages and limitations.	
OR		
(b)	Explain the principles and applications of electrochemical biosensors, focusing on their role in disease diagnostics.	
Q4	(10 marks)	CO5
(a)	Describe the lateral flow immunoassay (LFIA) technique and explain its role in rapid medical diagnostics. Also, discuss the Google Smart Lens technology in detail.	
OR		
(b)	Explain the importance of biosafety and ethical considerations in the commercialization of biosensors. What regulatory guidelines ensure the safe use of biosensors in healthcare and industry?	
Q5	(10 marks)	CO3
(a)	Discuss the role of transducers in biosensors, comparing different types such as electrochemical, calorimetric, potentiometric, amperometric, and piezoelectric biosensors.	
OR		
(b)	Evaluate the future prospects of biosensor technology. What are the emerging trends and applications in healthcare and industry?	

